

Roll No.

TCE-813

**B. TECH. (CIVIL ENGINEERING) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022**

HYDROPOWER ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) A run-of-river plant with an installed capacity 10,000 kW operates as a peak load station with a weekly load factor of 20%. What will be the minimum flow in the river so that the station may serve as the base load station ? Operating head = 15 m and plant efficiency = 80%. Calculate the daily load factor of the plant if the stream flow is 15 cumecs.

10 Marks (CO1)

OR

- (b) A four way is to design to meet out the pondage requirement of a runoff river scheme. The daily flow in the river is $20 \text{ m}^3/\text{s}$. The plant is to be used as 15 hrs peaking station. Predict the pondage factor and storage capacity of four ways.

10 Marks (CO1)

P. T. O.

2. (a) Discuss the various constraints in hydropower development with suitable examples. 10 Marks (CO1)

OR

- (b) A runoff river plant on a stream have a inflow of $20 \text{ m}^3/\text{s}$ and net head of 30 m with provision of pondage to meet daily peak demand with a load factor of 60%. Determine the power generation capacity of a plant at 80% of overall efficiency. The plant runs at a peak station for 3 hrs and balance period in a day for average load. Predict the amount of pondage needed. 10 Marks (CO1)
3. (a) Explain the Diversion Canal Plant and its different layouts with a neat sketch. 10 Marks (CO2)

OR

- (b) A closed cycle plant in Austria with a gross head of 350 m, has a headrace tunnel 4 m dia and 700 m long. The powerhouse discharge directly in the lower reservoir. The flow velocity is 6.5 m/s and the friction factor $f = 0.018$. If the overall efficiencies of pumping and generation are 85% and 88% respectively, determine the plant efficiency. 10 Marks (CO2)
4. (a) The turbine is to operate under a head of 24 m at 200 r.p.m. The discharge is $8.5 \text{ m}^3/\text{s}$. If the overall efficiency is 88%, determine :
- 10 Marks (CO2)
- (i) Power generated
 - (ii) Specific speed of turbine
 - (iii) Type of turbine

(3)

OR

- (b) A turbine develops 6620 kW under a head of 20 m at 130 r.p.m. Calculate the specific speed of the turbine and state the type of turbine.

10 Marks (CO2)

5. (a) In a hydroelectric station, water is available at the rate of $175 \text{ m}^3/\text{s}$ under a head of 18 m. The turbine runs at a speed of 150 r.p.m. with overall efficiency of 82%. Determine the number of turbine required if they have the maximum specific speed of 460.

10 Marks (CO2)

OR

- (b) Calculate the specific speed of a turbine and suggest the type of turbine required for a river having a discharge of 240 L/s with an available head of 45 m. Take the efficiency of turbine = 82% and speed = 450 r.p.m.

10 Marks (CO2)

60

Roll No.

TCE-801

B. TECH. (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

EARTHQUAKE RESISTANT DESIGN OF BUILDINGS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) A body of mass 2 kg is attached at free end of a cantilever beam of length 1.5 m having cross section (200 mm × 400 mm) and Young's modulus of 200 GPa. Find the natural frequency of system in hertz.

10 Marks (CO1)

OR

- (b) Derive the expression for response for a mass underdamped free vibration in single degree of freedom system.

10 Marks (CO1)

2. (a) A vibratory body of mass 150 kg supported on springs of effective stiffness 1050 kN/m has rotating unbalanced force of 525 N at a speed of 6000 rpm. If damping ratio is 0.3, find the amplitude caused by unbalanced force and its phase angle.

10 Marks (CO1)

P. T. O.

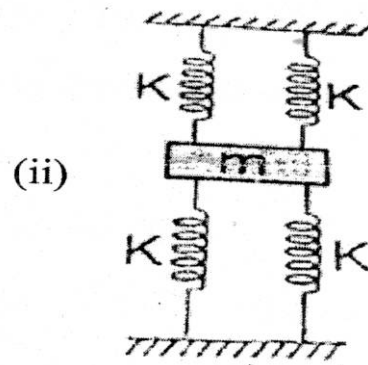
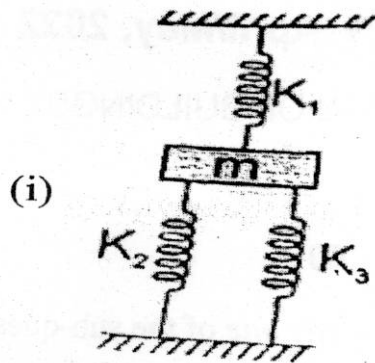
(2)

TCE-801

OR

- (b) How would you find the ration of transmissibility of forces between a machine (having rotating mass) and its foundation ? 10 Marks (CO1)
3. (a) Find the effective stiffness of the springs attached in system :

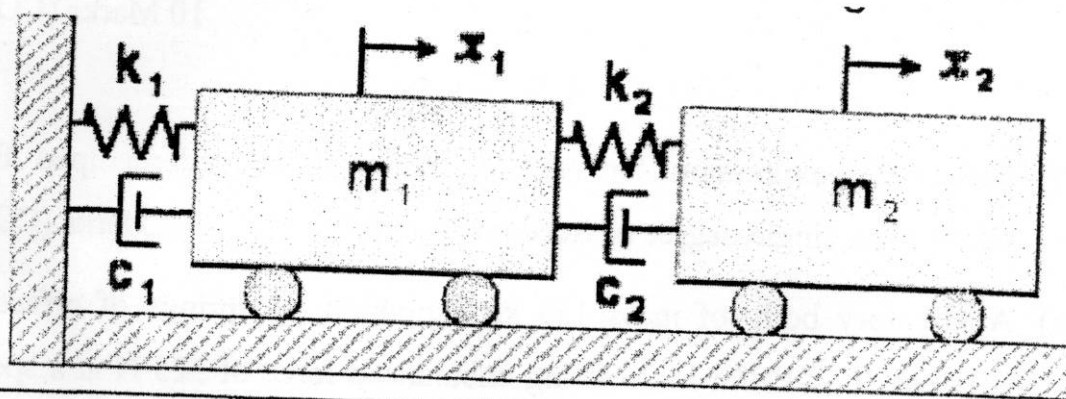
10 Marks (CO2)



OR

- (b) Write down the equation of motion for the given multidegree of freedom system.

10 Marks (CO2)



4. (a) How would you classify the different types of earthquakes on the basis of their occurrence ?

10 Marks (CO3)

(3)

OR

- (b) How would compare the effects and observations of earthquakes on modified Mercalli intensity scale and related the intensity scale with Richter scale. 10 Marks (CO3)

5. (a) How would you use the response spectrum for study of earthquake response on structures ? 10 Marks (CO3)

OR

- (b) What is the main ideal behind the working of a seismograph and how it can be used for design of earthquake resistant structures ?

10 Marks (CO3)